

# Automated Retail Analytics

## Sample Report — Retail Sales & Customer Analysis

### Data Source

E-commerce Sales & Customer Data (2009–2011)

### Analyses Included

- ① Data Cleaning & Quality Assurance
- ② Sales Trend Analysis (Monthly / Weekly / YoY)
- ③ Product Performance & Pareto Analysis
- ④ Customer Segmentation (RFM)

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[Available for freelance data analysis engagements](#)

Background & Objective

An e-commerce retailer held two years of sales and customer data but lacked the tools to turn it into decisions. This report structures the data across three lenses — revenue trends, product performance, and customer behaviour — and surfaces next actions.

£17.7M

Total Revenue

2009–2011

5,878

Customers Analysed

After data cleaning

Top 20%

of products drive 80%

of revenue (Pareto)

Key Findings (3)

- 1

Revenue shows strong November seasonality, consistent across both years.  
→ Stock and staffing must be prepared by mid-September at the latest.
- 2

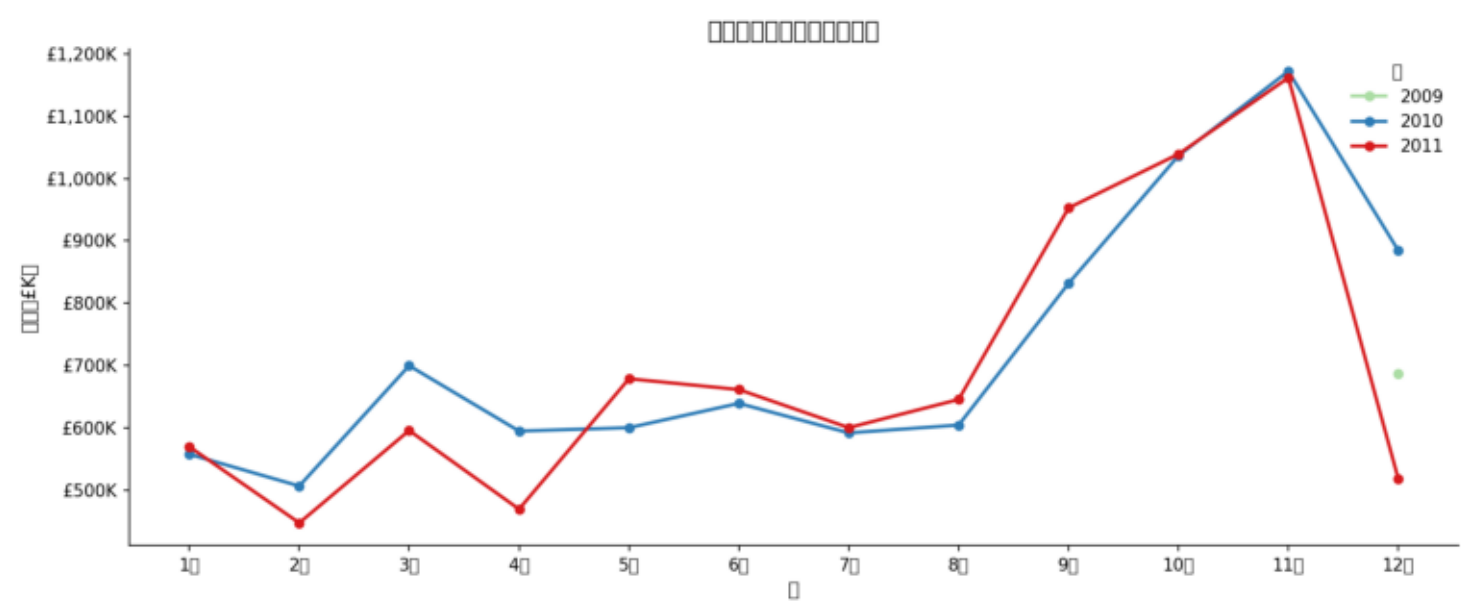
The top 20% of products generate the majority of revenue (Pareto principle).  
→ Preventing stockouts on hero SKUs is the highest-leverage inventory action.
- 3

Champion customers are a small share of the base but contribute a disproportionate share of revenue. Retaining them delivers the best return on any retention spend.

Dataset Snapshot

|                  |                               |
|------------------|-------------------------------|
| Analysis Period  | December 2009 – December 2011 |
| Transaction rows | ~805,000 (after cleaning)     |
| Unique customers | 5,878                         |
| Unique products  | 5,283 SKUs                    |
| Markets          | UK-centric · 40+ countries    |

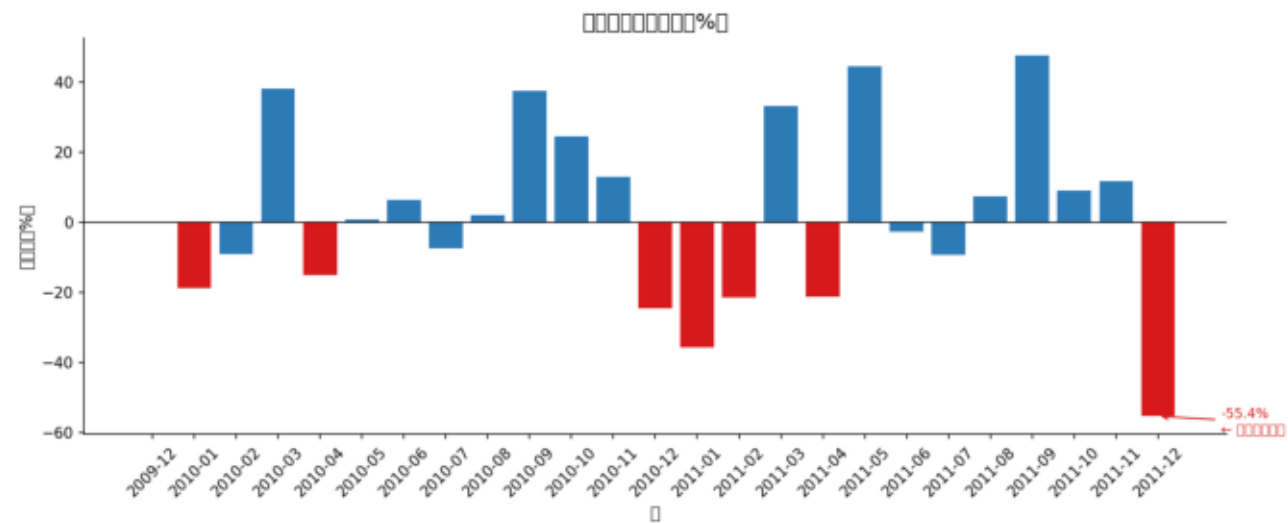
Monthly Revenue — Year-on-Year Comparison



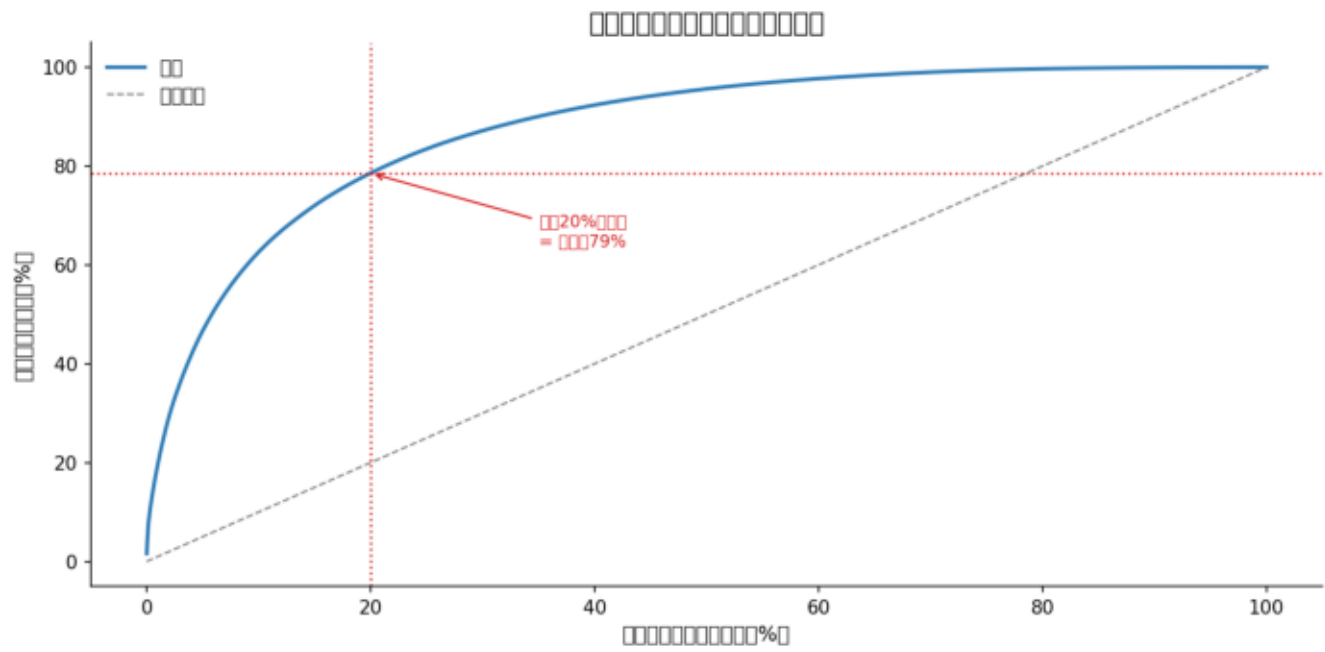
Analysis Commentary

- ▶ A November revenue spike is confirmed for two consecutive years, driven by holiday-season demand.
- ▶ Overall revenue grew year-on-year in 2011, confirming the underlying business growth trajectory.
- Inventory and staffing prep must begin by mid-September at the latest. Acting early on holiday demand is the single biggest revenue lever.

Month-over-Month Revenue Change (%)



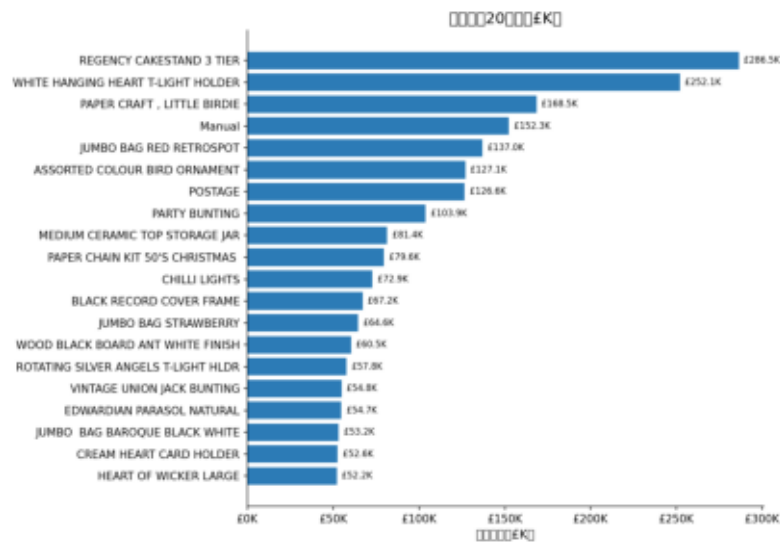
Pareto Analysis — Revenue Concentration



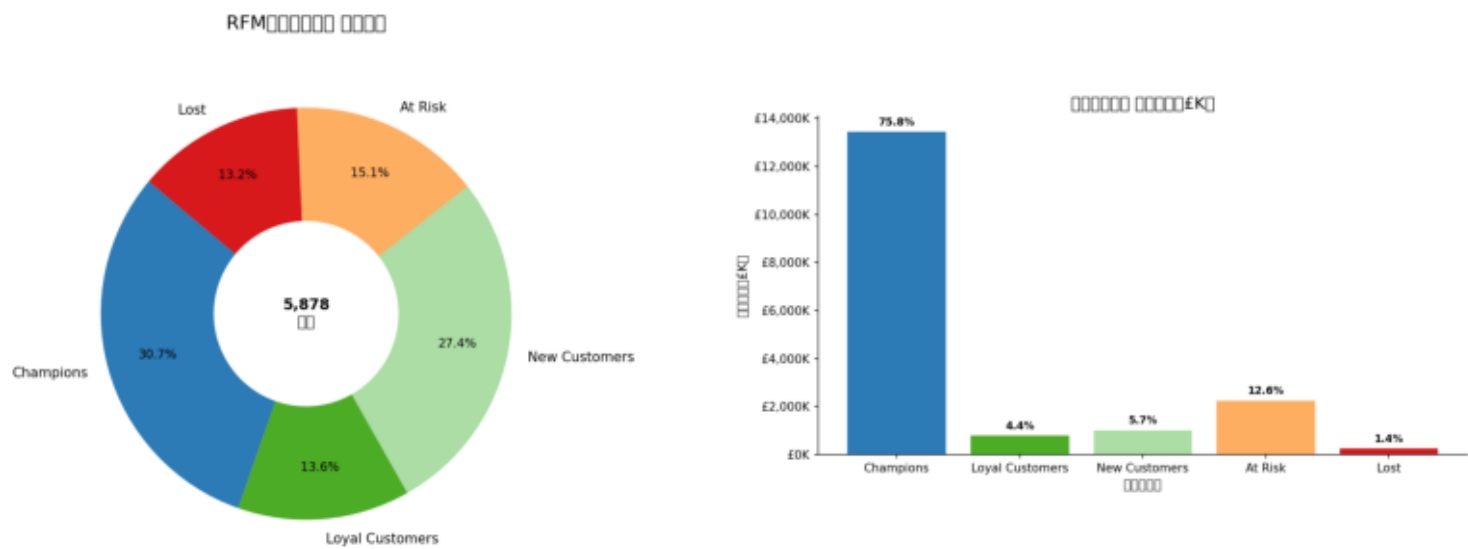
Analysis Commentary

- ▶ The Pareto principle holds: the top 20% of SKUs account for the large majority of revenue.
- ▶ High-order-frequency products with low revenue share are candidates for a modest price increase.
- Zero stockouts on the top 10 SKUs is the single most impactful supply-chain target — these items alone represent roughly 40% of total revenue.

Top 20 Products by Revenue



Segment Distribution — Customer Count & Revenue Share



Recommended Actions by Segment

|                 |                                 |                                           |
|-----------------|---------------------------------|-------------------------------------------|
| Champions       | Recent · Frequent · High-value  | VIP perks, early access to new products   |
| Loyal Customers | Repeat buyers, high engagement  | Loyalty programme, subscription discounts |
| At Risk         | Past high-value, recently quiet | Limited-time coupon within 30 days        |
| New Customers   | First or second purchase        | 7-day post-purchase follow-up email       |
| Lost            | Long-lapsed, no recent activity | Win-back campaign or retire from list     |

➔ Champions are ~31% of customers but drive ~76% of revenue. Protecting this group has a higher ROI than acquiring new customers.

RFM Score Heatmap by Segment



Average R (Recency), F (Frequency), and M (Monetary) scores per segment. Champions score high across all three dimensions; Lost customers score low across all three.

Seasonal Decomposition (Trend · Seasonality · Residual)

Revenue is decomposed into long-term trend, seasonal pattern, and residual noise. The upward trend persists after removing seasonality — confirming genuine underlying growth.



## Methodology

This report was produced using industry-standard analytical methods. Results are presented in plain language — no technical background required.

### ■ Data Cleaning

Removed cancellations, missing customer IDs, and invalid prices. Only reliable records enter the analysis.

### ■ Trend Analysis

Monthly and weekly aggregation with year-on-year comparison. Seasonal decomposition isolates structural growth.

### ■ Pareto Analysis

Visualises revenue concentration. Identifies which products and customers deserve the most focus.

### ■ RFM Segmentation

Scores customers on Recency, Frequency, and Monetary value, then classifies them into 5 actionable groups.

## Enquiries & Engagements

Active data scientist. Available for freelance analysis projects.

- Send your Excel or CSV — that's all we need to get started
- Delivered as a branded PDF or Excel workbook
- Plain-language commentary with every chart
- Happy to scope open-ended or unclear briefs

Get in touch — first conversation is free.

This is a sample report. All analyses are customised to your actual business data and objectives.